Diabetes Prediction Using Machine Learning

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***Abstract*—Roughly 27.6% of diabetic adults in the United States remain undiagnosed, making scalable, low-cost screening an important area of public health research. We trained and compared four classification models on the 2015 CDC Behavioral Risk Factor Surveillance System (BRFSS) to predict type 2 diabetes diagnosis from self-reported survey responses. The dataset required careful preprocessing, including feature selection drawn from published T2D research, per-variable handling of BRFSS administrative codes, and collapsing a three-class target into a binary one due to severe pre-diabetes underrepresentation. The four model families evaluated were Logistic Regression, Linear SVM, Kernel SVM, and a PyTorch MLP neural network trained across three optimizer configurations.**

All models were tested on a held-out set of 50,736 samples at the natural 84/16 class distribution. Since a missed diabetes diagnosis carries far greater clinical cost than a false positive, recall was treated as the most important per-class metric, with ROC-AUC used for model comparison. ROC-AUC scores ranged from 0.817 to 0.825 across all models. The MLP with a tuned positive class weight achieved the highest recall at 0.869 and matched the top AUC of 0.8245. BMI, age, and self-rated general health were the strongest predictors across models. The narrow gap between linear and non-linear models points to self-report noise as the main constraint on classification performance, rather than model capacity.

***Index Terms*—diabetes, machine learning, classification, support vector machines, neural networks, BRFSS, public health**

I. Introduction

Diabetes has posed a persistent public health challenge across the United States for decades, and increasingly so in developing countries. Diabetes symptoms are debilitating and can impart significant financial strain on both individuals and the broader healthcare system.

A late diagnosis of diabetes can result in life-changing complications, including increased cardiovascular disease, neuropathy, retinopathy, and fatty liver disease. Neuropathy and retinopathy, in their worst, untreated forms, can result in amputation and blindness. The CDC estimates that 40.1 million Americans have diagnosed or undiagnosed diabetes, with an additional 115.2 million adults living with prediabetes — and 27.6% of diabetic adults remain undiagnosed [1].

The recent advent of GLP-1 receptor agonists as an effective treatment for type 2 diabetes makes early detection and intervention more crucial than ever, as earlier diagnosis enables earlier access to these therapies. This motivates the use of population-level survey data for ML-assisted screening, and we analyze the CDC's Behavioral Risk Factor Surveillance System [2] dataset with a suite of classification models.

The BRFSS dataset contains survey responses from over 400,000 Americans each year. The dataset is nationally representative but inherently noisy due to its self-reported nature, and it is heavily class-imbalanced, with a roughly 85/15 split between non-diabetics and diabetics.

The objective of this work is threefold: build and compare a suite of classification models to predict type 2 diabetes diagnosis, identify which features carry the strongest predictive signal, and determine which models perform best with respect to interpretability, performance, and generalization. While the dataset admits a three-class target distinguishing no diabetes, pre-diabetes, and diabetes, we limit modeling to a binary formulation — collapsing pre-diabetes and diabetes into a single positive class — due to the severe class imbalance the three-class target introduces. We explore Logistic Regression, Linear SVM, Kernel SVM, and a variety of Artificial Neural Networks, using systematic hyperparameter tuning to derive the best performance possible from each. Model performance is evaluated using ROC-AUC, recall, and accuracy. Since the work is clinical in nature, a model that produces fewer false negatives is treated as providing higher quality predictions.

The remainder of this report covers data selection and preprocessing, exploratory data analysis, the modeling process and hyperparameter tuning, results and model comparison, and conclusions with directions for future work.

II. Data and Preprocessing

A. Data Overview

The data is from the 2015 BRFSS file. The dataset consists of 441,456 rows with 330 different columns. We had initially considered using data from multiple years to collate the data into an even larger set, but there were many variations in the survey. We could not confidently say that there was sufficient consistency across years to concatenate the datasets.

B. Preprocessing

1) Feature Selection: Feature selection is grounded in published diabetes research. The CDC's BRFSS has been used extensively in T2D prediction studies, which consistently identify the following domains as predictive: metabolic markers (BMI, blood pressure, cholesterol), lifestyle factors (physical activity, diet, smoking, alcohol), chronic comorbidities (stroke, heart disease), healthcare access, self-reported health status, and demographics (age, sex, income, education).

We retain the target DIABETE3 and 21 clinically recommended features. All administrative columns are dropped. The selected features and their rationale are listed in Table I. The features are quantitative, qualitative, and behavioral in nature. These features reveal information concerning the quantitative measures of the body's health, including blood pressure, cholesterol, and BMI, as well as information that is qualitative and behavioral, including frequency of health screenings, the type of diet, and effective mobility.

2) Missing Data & Numerical Transformation: The data included a small number of truly missing observations, but review of the data format book reveals administrative coding used which obfuscated missing data (7, 9, 99, etc). We also normalized all BMI values by dividing by 100, but some of these BMI values were high. We decided to remove any BMI above 100, and also removed any rows with a BMI of 99.99 as those were refusal to answer.

3) Target Variable Construction: We converted all target variables to a standard encoding format; 0, 1, and 2 for non-diabetic, pre-diabetic, and diabetic, respectively. After exploring the data balance, we determined the best path forward was to collapse pre-diabetes and diabetes into a single diabetes target variable. Pre-diabetes had very minimal representation in the dataset.

4) Train/Test Split & Class Imbalance Strategy: We used a simple 80/20 train and test split. We had 202,944 training samples and 50,736 testing samples. The target class ratios were preserved.

The natural distribution of the class was 84.2% negative and 15.8% positive for diabetes. The models required unique handling of the data imbalance:

- **Logistic Regression:** `class_weight='balanced'` was applied.
- **Linear SVM:** Trained on an undersampled balanced training set of 63,964 samples (50/50 class ratio), as the loss is sensitive to imbalance and does not natively support class weighting in the same manner.
- **Kernel SVM (RBF):** Also trained on the balanced set; additionally subsampled to 20,000 rows to satisfy memory constraints imposed by the $n \times n$ kernel matrix.
- **Neural Network (MLP):** A `pos_weight` parameter in the `BCEWithLogitsLoss` function was used to upweight the positive class during training. This value was tuned as a hyperparameter across several iterations of the model.

5) Scaling: Applied the `StandardScaler()` function on training set only. Applied identically across splits to prevent any data leakage.

III. Exploratory Data Analysis

We performed exploratory data analysis to uncover more information regarding class distribution, target-feature and intra-feature correlation, and feature importance through a coefficient path.

TABLE I
SELECTED BRFSS FEATURES AND CLINICAL RATIONALE

BRFSS Column	**Feature**	**Rationale**
DIABETE3	Target	Diabetes diagnosis (3-class)
_RFHYPE5	HighBP	High blood pressure is a leading T2D comorbidity
TOLDHI2	HighChol	High cholesterol is an established T2D predictor
_CHOLCHK	CholCheck	Recent screening indicates healthcare engagement and earlier detection
_BMI5	BMI	Obesity is the single strongest modifiable risk factor
SMOKE100	Smoker	Smoking increases insulin resistance and fat accumulation
CVDSTRK3	Stroke	Stroke and T2D share vascular and inflammatory risk pathways
_MICHD	HeartDisease	Strongly correlated with undiagnosed or poorly controlled T2D
_TOTINDA	PhysActivity	Sedentary behavior drives insulin resistance; exercise is a primary prevention tool
_FRTLT1	Fruits	Low fruit intake is a diet quality marker linked to T2D risk
_VEGLT1	Veggies	Low vegetable intake correlates with high glycemic index dietary patterns
_RFDRHV5	HvyAlcohol	Heavy drinking disrupts glucose regulation and liver insulin sensitivity
HLTHPLN1	AnyHealthcare	Insurance coverage affects screening rates and diagnosis likelihood
MEDCOST	NoDocbcCost	Social determinant of late T2D diagnosis
GENHLTH	GenHlth	Self-rated health is a robust predictor of chronic disease
MENTHLTH	MentHlth	Depression and anxiety are bidirectionally linked to T2D
PHYSHLTH	PhysHlth	Poor physical health reflects overall likelihood of chronic disease
DIFFWALK	DiffWalk	Mobility difficulty is associated with obesity, neuropathy, and advanced T2D
SEX	Sex	T2D prevalence varies by sex
_AGEG5YR	Age	Age is the strongest non-modifiable T2D risk factor; risk accelerates after 45
EDUCA	Education	Education shapes health literacy, diet, and preventive care access
INCOME2	Income	Income determines access to healthy food, care, and physical activity infrastructure

A. Class Distribution

We have found the class distribution to be heavily skewed towards non-diabetics, naturally. We also illustrate both the tri-class and binary-class target breakdowns in Fig. 1.

On the righthand side, we show the 84/16% split between binary target classes. On the lefthand side, we show that pre-diabetes is only represented in just under 2% of the data. Training to classify such a small portion of the dataset would result in very poor predictions, and thus we determined it was not achievable given the current dataset.

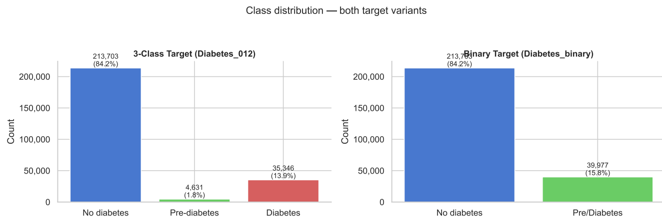

Fig. 1. Class distribution of the binary and three-class diabetes targets.

B. Feature Visualizations

We evaluated the distribution of BMI across both target groups: non-diabetic and diabetic. The chart shows a clear shift to higher BMI among diabetic individuals as compared to non-diabetic individuals.

There are vertical dashed lines indicating the different regions of BMI status; underweight, normal, overweight, and obese. The majority of the observations range between a BMI of 18 and 40, though you can see a strong right-tailed skewness.

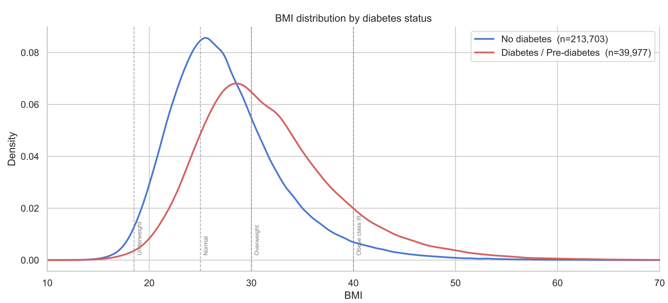

Fig. 2. BMI distribution by diabetes status.

We also calculated diabetes rate by feature value for all 21 features. Here we share three:

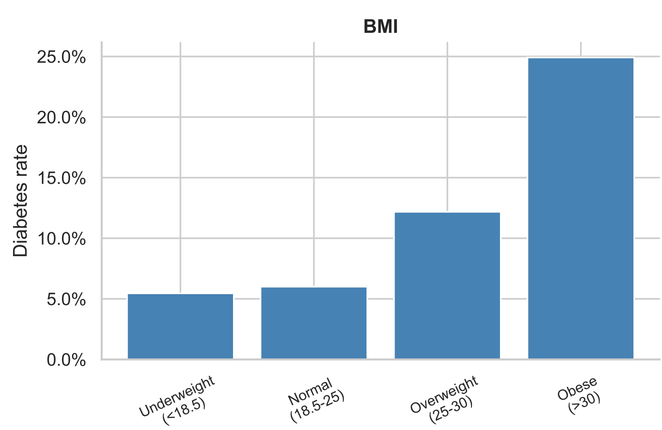

Fig. 3. Diabetes rate by BMI. The chart makes it clear; underweight and normal weight individuals have similar rates of diabetes, but overweight doubles the rate of diabetes - and obese is five times the rate of diabetes as normal weight.

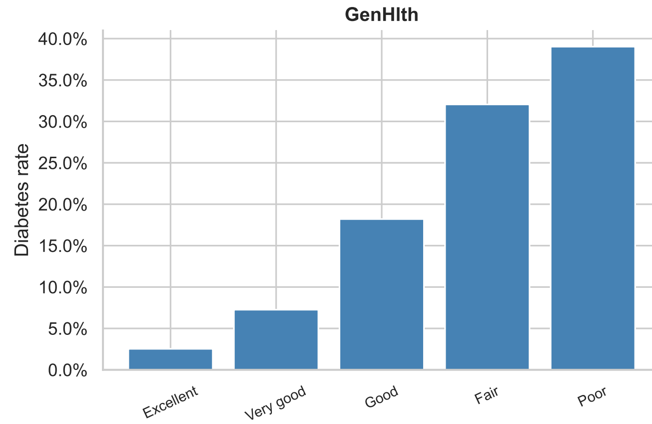

Fig. 4. Diabetes rate by General Overall Health. There is a clear inverse relation between general health and rate of diabetes.

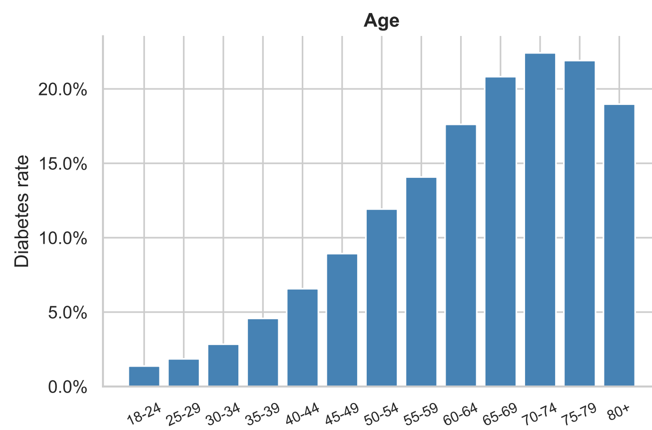

Fig. 5. Diabetes rate by Age. Increasing age coincides with increased rate of diabetes, except for the oldest age group.

C. Correlations

We calculated Pearson correlations both among features and between each feature and the binary diabetes target. The full correlation matrix is provided in Fig. 16 in the Appendix. The strongest positive correlates with the target are BMI, GenHlth, PhysHlth, DiffWalk, and Age. Income and Education show the weakest associations. Multicollinearity among features is low, confirming that the selected feature set carries largely independent predictive signal.

D. L1 Regularization Coefficient Path

To assess feature robustness under increasing regularization, we swept the L1 penalty strength across logistic regression models and recorded each feature's coefficient as it was driven toward zero. Features whose coefficients survive the strongest regularization are the most robust linear predictors of diabetes. BMI, Age, and GenHlth emerge as the most robust features, while Income and Education are eliminated earliest. The full coefficient path is provided in Fig. 17 in the Appendix.

IV. Modeling

We used four different classification models to predict the binary target class. We found that many of the models performed similarly, but the MLP neural network had the best performance.

A. Logistic Regression

The logistic regression model worked very well as a lightweight model; it was quick to train and performed as well as the linear SVM and nearly as well as the kernel SVM at a fraction of the training time. Since the model has in-built methods for handling data imbalance, we were able to use the standard dataset without any modifications. We used the Limited-memory Broyden-Fletcher-Goldfarb-Shanno `lbfgs` solver as our optimizer, L2 regularization to reduce overfitting, and set `class_weight = 'balanced'` to account for the imbalanced dataset. We employed Ridge regression as we had already tested Lasso regression on these features with the L1 Coefficient walk in Fig. 17 and found that there were low levels of multicollinearity among the subset.

After setting the base parameters, we performed grid search cross-validation over five folds of the data. We tested $C \in \{0.001 - 10\}$, and selected our best C through ROC-AUC scores. We then created a confusion matrix for the binary target variable, and identified our most informative features.

B. Linear and Kernel Support Vector Machines

The linear SVM model was trained on the 50/50 balanced dataset. We used a `CalibratedClassifierCV` to generate the probability outputs required to perform classification. We performed hyperparameter tuning over five folds to find our optimal $C \in \{0.001 - 10\}$ and used ROC-AUC scores to determine the best C. We once again created a confusion matrix to evaluate performance.

The kernel SVM model was used to find non-linear decision boundaries for the classification, which allowed us to calculate probability estimates. An `N x N` kernel matrix would be infeasible with 63K rows, so we used a 20K row subsample of the 50/50 balanced dataset. We performed a grid search over $C \in 0.1, 1, 10$ and $\gamma \in scale, 0.01, 0.1$. We again used both ROC-AUC and a confusion matrix to evaluate the performance of the classifier.

C. Artificial Neural Network

The MLP-architecture for the artificial neural network was used as our base. We performed three iterations of modeling, one with Adam optimization, one with AdamW optimization, and one with tuned `pos_weight`. We tested a variety of architecture structures, including (64, 32), (128, 64), and (128, 64, 32) hidden layers, with ReLU activations and dropout of 0.3 to prevent overfitting.

We used `BCEWithLogitsLoss` to predict the class. We track F2 scores throughout the process to give proper consideration to false negatives. Each iteration tracked best performance and used a 90/10 split on train/test data for the cross-validation portion of the process.

1) Adam: We used a learning rate of 1e-3 for the first iteration. We also used a natural pos_weight (5.33). Leveraging `ReduceLROnPlateau` helped to avoid getting stuck in a suboptimal local minima of loss. We also use early stopping with a patience of 15.

2) AdamW: For the shift to the AdamW optimizer, we used a learning rate of 5e-4. This decouples the weight decay from adaptive gradient scaling. We still used ROC-AUC and the confusion matrix to evaluate the performance.

3) AdamW + tuned pos_weight: The last iteration of the artificial neural network training was to implement a tuned `pos_weight` to minimize the number of false negatives while maintaining a reasonable degree of accuracy for the true negatives, true positives, and false positives. We tracked the F2 score as our optimizer for performance during the training process. We searched across the following values: `pos_weight` $\in \{2.0, 5.33, 8.0, 10.66, 16.0\}$. We also modified the architechture for this iteration, using (64,32), (128,64), (128,64,32) over a `weight_decay` $\in \{1e{-}4, 1e{-}3, 1e{-}2\}$.

V. Results

In this section we will review the results from the analyses of the four different models that we trained and tested.

A. Evaluation Protocol

All models are evaluated on the same held-out test set (50K rows) with the natural distribution. No steps were taken to correct the imbalance for the test dataset. We collected metrics for Accuracy, Precision, Recall, F1, ROC-AUC, the Confusion Matrix, and the F2 scores for the previously mentioned ANNs.

The ROC-AUC score was used as the primary metric because it is threshold-independent and robust to class imbalance. Recall was the secondary metric because it aligned with the clinical goals of the classifer; namely, that a false negative carries significantly more cost than a false positive for a patient. Accuracy was used in a limited manner, as the metric could be misleading. For example, if the classifier predicted all data points as negative for diabetes then we would achieve nearly 85% accuracy.

B. Logistic Regression

Logistic regression provided adequate results. Strong regularization provided the best results, so we selected a value of 0.001 for C. The model scored 0.8178 for cross-validated training AUC. On the test set, the model achieved 0.8177 for ROC-AUC, 0.731 for accuracy, 0.761 for recall, 0.341 for precision, and 0.471 for the F1 score.

As seen in Fig. 6, the model correctly identified 76% of diabetic cases, with a 73% overall accuracy rate. The low precision is expected given the imbalanced dataset. At high recall thresholds, false positives are unavoidable. Setting `class_weight = 'balanced'` compensated well for the imbalance, though. The strong L2 regularization required suggests we have an overfitting risk without it.

Shown in Fig. 7, General health, BMI, age, and high blood pressure all had the largest positive coefficients. Physical activity and income carried the largest negative coefficients.

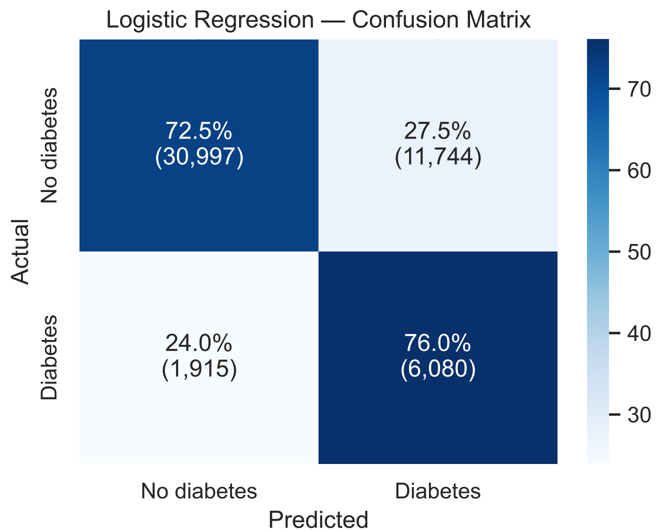

Fig. 6. Confusion matrix for the Logistic Regression model. Adequate performance.

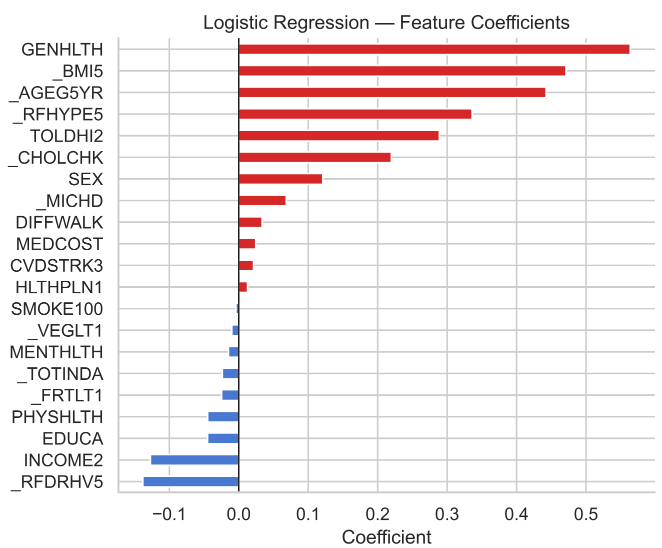

Fig. 7. Feature coefficients for the Logistic Regression model.

C. Linear SVM

Linear SVM performed nearly identically to Logistic Regression. We used a C of 0.01, so less regularization, and achieved a score of 0.8157 for cross-validated AUC. The accuracy scored 0.731, recall scored 0.760, precision scored 0.341, and the model achieved 0.471 on the F1 score.

Comparing the confusion matrices (Fig. 6, Fig. 8) shows how aligned the two models were in their classifications. Both models converge to nearly the same linear decision boundary, which confirms that the problem is very likely linearly separable.

Linear SVM was trained on the 50/50 undersampled set with 63,964 samples. There was no meaningful performance disadvantage despite using one-fifth of the training samples.

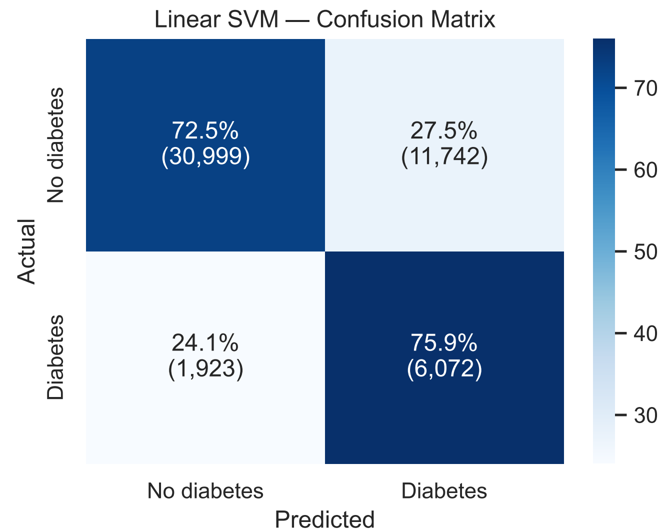

Fig. 8. Confusion matrix for the Linear SVM model. Nearly identical performance to Logistic Regression.

In a practical setting, we would certainly employ this model over Logistic Regression.

D. Kernel SVM

The kernel SVM model used a C of 1, considerably less regularization. The model achieved a score of 0.8185 on the cross-validated AUC, and a score of 0.8205 on the test set ROC-AUC. The model scored 0.701, 0.812, 0.322, and 0.461 for accuracy, recall, precision, and F1, respectively.

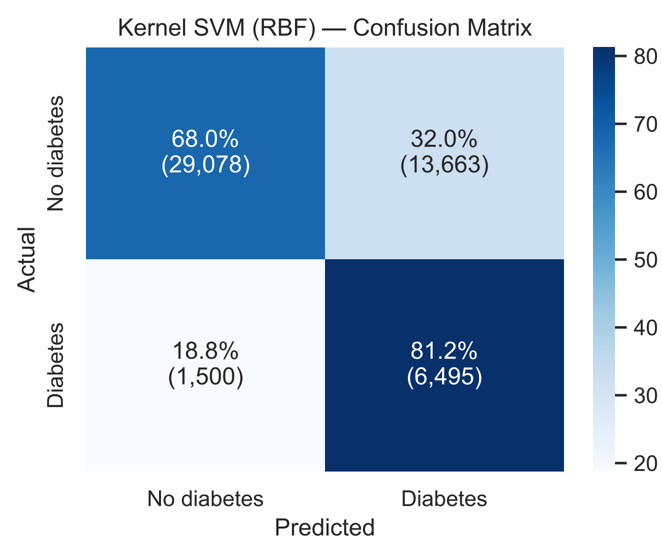

Fig. 9. Confusion matrix for the Kernel SVM model. Reduced false negative predictions at the cost of increased false positives.

This model had the highest recall of any model family tested, correctly capturing 81% of diabetes cases. The gain for the area under the curve was modest, which again suggests the data's structure is linear. We traded off lower accuracy for more aggressive positive class predictions.

The model was trained on just 20,000 rows, a subsample selected due to memory constraints. The AUC is still slightly higher than Linear SVM, despite this limited dataset.

E. Artificial Neural Network

1) Adam: The first iteration of ANN MLP used a learning rate of 1e-3 and a natural `pos_weight` of 5.35 before testing a variety of architechtures. We found that (128, 64, 32) layers with a weight decay of 0.0001 performed best.

The model scored 0.8242 on the test ROC-AUC, 0.715 on accuracy, 0.795 on recall, 0.331 on precision, and 0.468 on the F1 score.

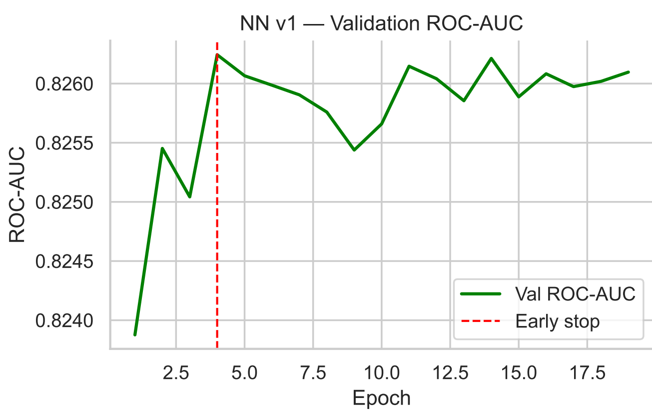

Fig. 10. Validation AUC for the first iteration of the ANN.

This immediately performed with the highest AUC of the four model families. The time required to train the neural network was comparable to the kernel SVM once we implemented early stoppage.

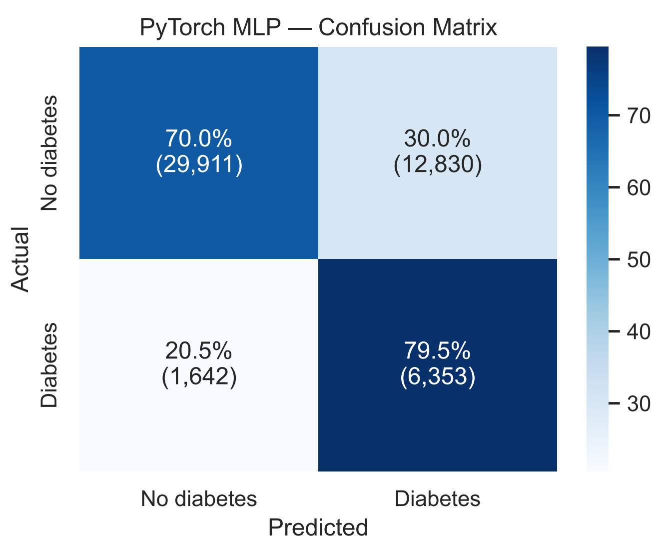

Fig. 11. Confusion matrix for the first iteration of the ANN.

The model performed just below the kernel SVM in recall, trading that performance for better identification of true negatives.

2) AdamW: The second iteration of the MLP used a learning rate of 5e-4 with the same architecture and `pos_weight` as the previous. AdamW decouples the weight decay from adaptive gradient scaling. The model scored 0.8245 on the test ROC-AUC, 0.720 on accuracy, 0.786 on recall, 0.335 on precision, and 0.470 on the F1 score.

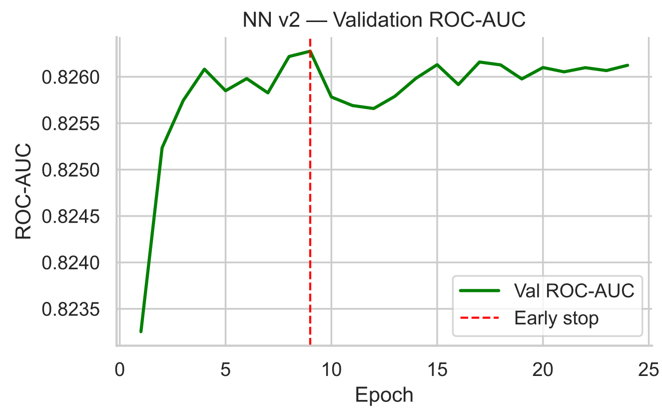

Fig. 12. Validation AUC for the second iteration of the ANN.

The result were very similar as the first iteration of the model (Fig. 13). We achieved slightly higher accuracy and precision with a modest recall decrease. We felt that we wanted to prioritize higher recall due to clinical objectives, so we moved forward with the third iteration of the model.

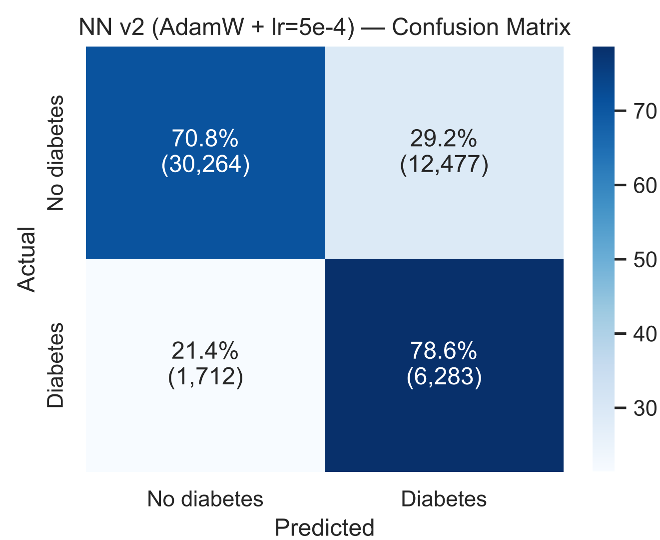

Fig. 13. Confusion matrix for the second iteration of the ANN.

3) AdamW + Tuned `pos_weight`: The last neural network leveraged a tuned `pos_weight` to incentivize higher recall. We trained it using a learning rate of 5e-4, and tuned across {2.0 - 16.0}, with the final landed `pos_weight` of 8.02, resulting in a greatly increased recall.

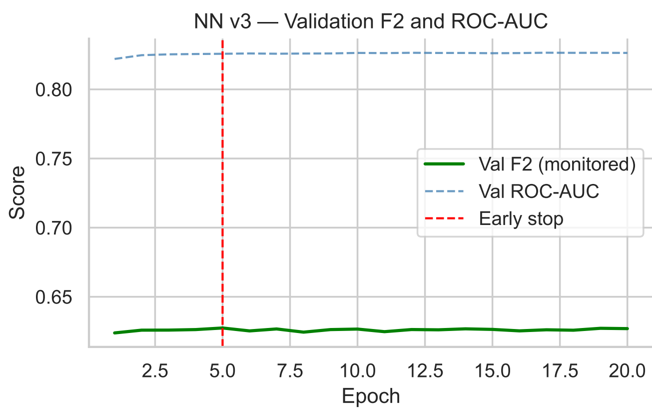

Fig. 14. Validation AUC and F2 for the third iteration of the ANN.

The model achieved a test ROC-AUC of 0.8245 and scored 0.650, 0.869, 0.294, and 0.439 for accuracy, recall, precision and the F1 score, respectively. Iteration three scored the same AUC as iteration two, but dramatically shifted the operating point. Recall rose by over 8% at the cost of 7% accuracy and 4% precision.

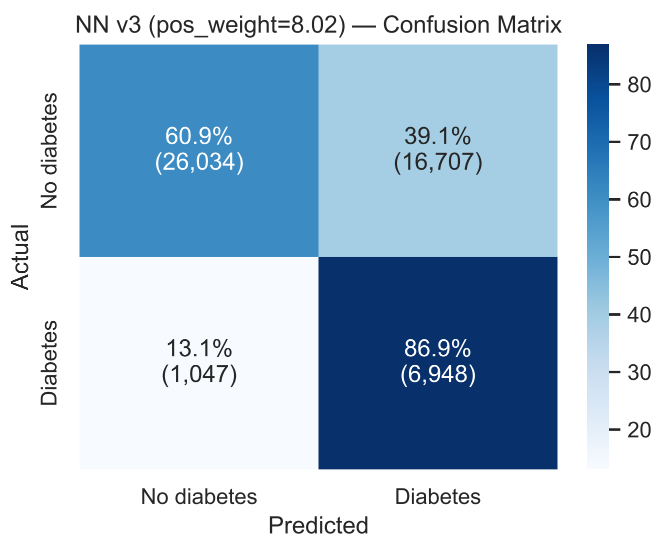

Fig. 15. Confusion matrix for the third iteration of the ANN.

This model had clinically favorable results; catching all but 13% of diabetic cases. This tuning was appropriate when the priority is to minimize missed diagnoses.

F. Model Comparison

Table II summarizes the performance of all six model configurations evaluated on the held-out test set of 50,736 samples, preserving the natural 84.2/15.8% class distribution.

Ultimately, the third iteration of the neural network, with the tuned `pos_weight`, was our best performing model. The model had the highest recall and the best AUC score, and it best aligned with our clinical motivations. The kernel SVM model also performed very well.

TABLE II
MODEL PERFORMANCE ON HELD-OUT TEST SET (DIABETES CLASS)

Model	Acc.	Prec.	Recall	F1	AUC
Logistic Regression	0.731	0.341	0.761	0.471	0.8177
Linear SVM	0.731	0.341	0.760	0.471	0.8175
Kernel SVM (RBF)	0.701	0.322	0.812	0.461	0.8205
NN v1 (Adam)	0.715	0.331	0.795	0.468	0.8242
NN v2 (AdamW)	0.720	0.335	0.786	0.470	0.8245
NN v3 (AdamW, pw=8.02)	0.650	0.294	**0.869**	0.439	**0.8245**

G. Discussion

All the models clustered rather tightly in the range of {0.817 - 0.825} for AUC. This suggests the ceiling is a data constraint rather than a model constraint. The self-reported nature of the survey features may introduce noise that limits the upper bounds of the model's performance.

There was a very small gap between linear vs non-linear models, which suggests that the relationship between features and class is largely linear. The non-linear models gained a small advantage at the cost of much greater compute requirements.

The recall vs precision tradeoff was dominant throughout this analysis. All models sacrificed in their precision {0.29 - 0.34} due to the imbalanced dataset. In a clinical setting, the meaningful decision to make concerns how aggressively one would want to bias toward recall. The third iteration of the neural network or the kernel SVM would be the clear winners if that were a primary focus.

Regarding the neural networks, much time was spent tuning and testing a variety of tunings and architectures, but there was little meaningful differences in the overall performance of the model. Only tuning the `pos_weight` heavily impacted performance, and that was an intentional tradeoff. There was no significant improvement to the AUC score. See Fig. 18 and 19 for ROC curves of all models.

VI. CONCLUSION

A. Summary of Findings

The four models perform comparably across the main metrics. MLP edges ahead of the other models on ROC-AUC. The non-linear gain is modest, and the MLP may not be as efficient when comparing compute requirements to classification performance.

B. Clinical Framing

Special considerations must be made for the purpose of this model; we are not simply predicting profits and losses, or blah and blah, but attempting to identify serious health risks in patients that could have massive impacts to their health outlook. Therefore, this necessitates focusing the model on recall, specifically, as that allows us to optimize the model against making false negative predictions.

Additionally, interpretability is a primary factor here, as well. The ability to measure feature importance allows for clinical recommendations to be made. This particular dataset has been well-researched, and new discoveries of contributing factors to diabetes aren't going to be found with this, but it is still very important to make appropriate decisions.

Kernel SVM performed noticeably better than the linear models with respect to the recall metric, and warrants further investigation. Since we were limited by our compute capabilities, we had to train on only a subset of the dataset, rather than the entire 2015 year of data. The main drawback of using a kernel SVM model is the lack of interpretability that it offers.

MLP benefitted significantly from flexibility in determining the loss; we were able to weight false negatives more heavily which supports good decisionmaking in a medical framework. We are much better off over-diagnosing than under-diagnosing when considering the relative risks. The neural network models also lack interpretability.

C. Limitations

The primary limitation of this analysis was the lack of computing power. Working on a local machine did not provide the necessary capabilities to train over very large datasets when using the kernel SVM or MLP models. Either running these models on more powerful hardware, locally, or using cloud computing to enhance our capabilities, would have allowed us to explore these models more. Acknowledging this, the dataset was still too small to warrant much more complex architecture on the neural networks.

Broadly, the models genuinely performed very similarly. That being said, it is rather apparent that the non-linear models outperformed the linear models by a noticeable margin, even on a small dataset like this. This points us toward other ledes to follow to further empower these classifiers, such as gathering a larger dataset. This would have required significant cleaning of the BRFSS datasets from other years, which adds variability due to changing questions and different framing of the same questions.

Additionally, this data is self-reported survey data, so noise and bias are inevitable. The data itself provides no causal inference. The dataset would provide much stronger causal information if combined with electronic health records from patients to inform of any adverse health events as well as annual lab and physical results. Some of these features can only be frequently self-reported, such as difficulty walking, or features indicating the intake of fruits and vegetables in the diet, but ultimately we would benefit greatly from matched electronic health records with each survey respondent.

D. Future Work

There are many avenues to explore after performing this analysis. Primarily, we would be eager to create a more sophisticated pipeline for the data that could intake many years of this survey data. The pipeline would necessarily have to handle the variety of the survey questions and any other administrative differences between years.

Another interesting addition could be the inclusion of feature interaction terms, or the expansion of the available features for the MLP neural networks. Though we are happy with the feature selection process for this analysis, we would find it interesting to explore interaction terms or simply keep more of the original features. This would naturally require more extensive data preprocessing and exploratory data analysis, as well as significantly more powerful compute capabilities.

Exploring tree-based models, such as Random Forest Classifiers or Gradient Boosted Classifiers could be interesting, as well. They would provide a lightweight model with very high interpretability, and could be an exciting analysis to conduct.

References

[1] Centers for Disease Control and Prevention, "National Diabetes Statistics Report," National Center for Chronic Disease Prevention and Health Promotion, Jan. 2026, accessed May 2026. [Online]. Available: https://www.cdc.gov/diabetes/php/data-research/index.html

[2] Centers for Disease Control and Prevention (CDC), "Behavioral Risk Factor Surveillance System Survey Data," U.S. Department of Health and Human Services, Centers for Disease Control and Prevention, Atlanta, Georgia, Tech. Rep., 2015. [Online]. Available: https://www.cdc.gov/brfss/annual_data/annual_data.htm

APPENDIX

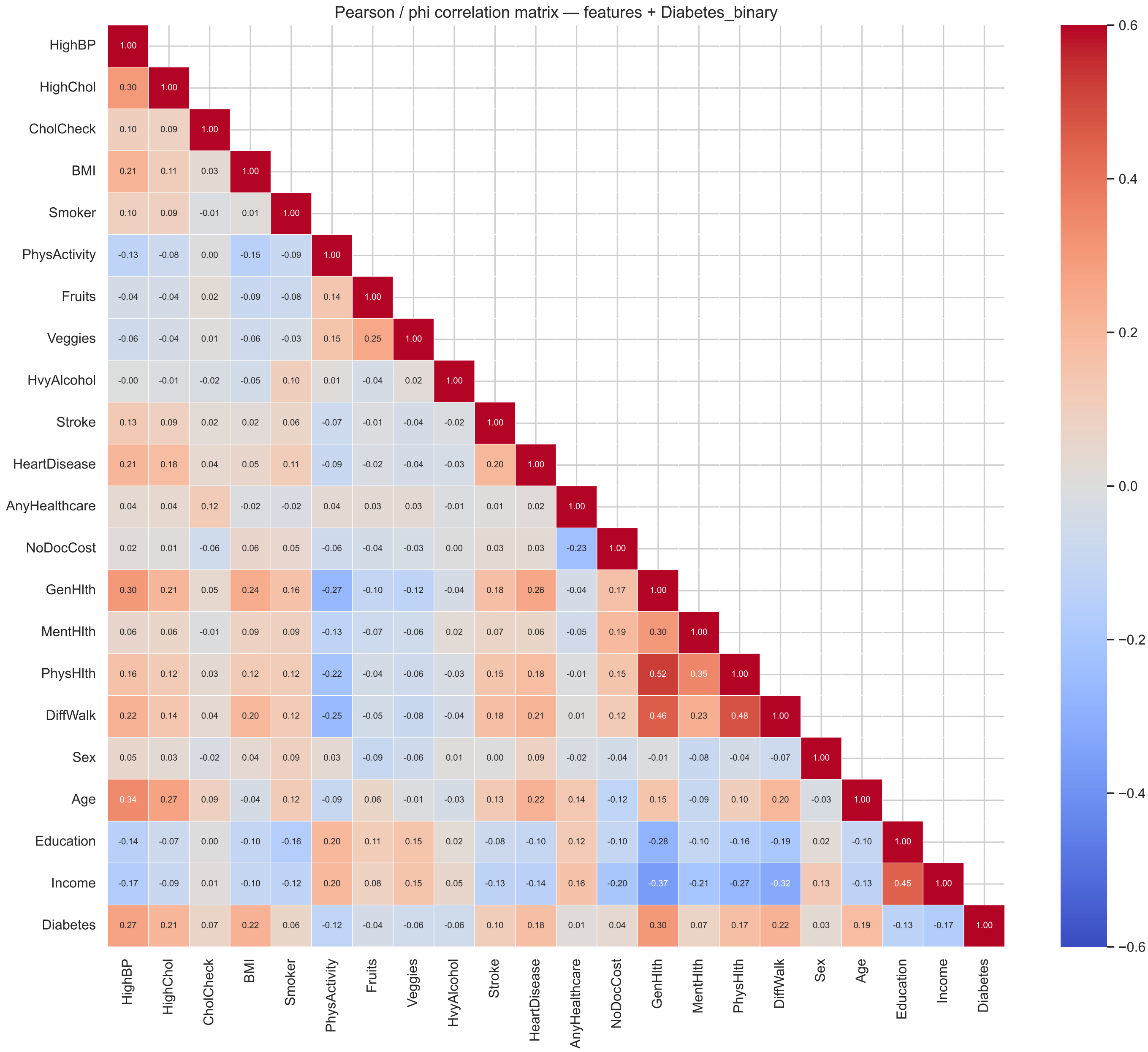

Fig. 16. Pearson/phi correlation matrix across all 21 features and the binary diabetes target. The lower triangle is shown. Positive associations appear in red; negative in blue.

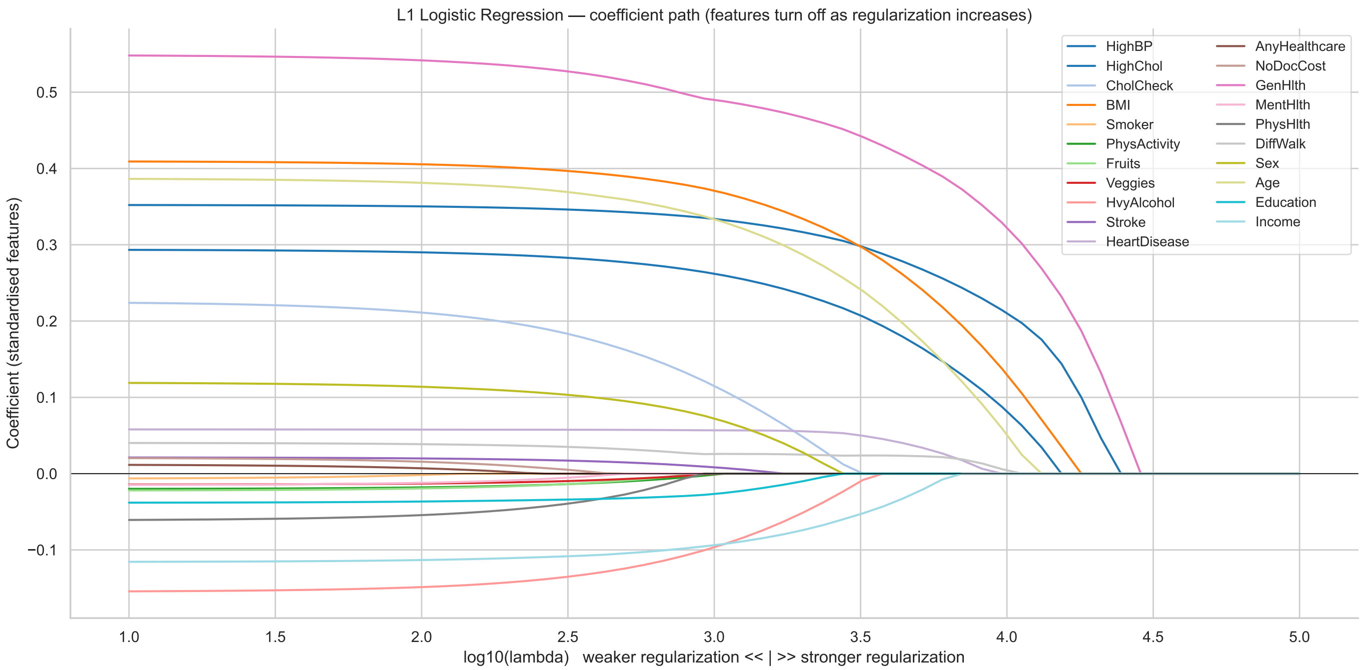

Fig. 17. L1 logistic regression coefficient path. Each line traces one feature's coefficient as regularization strength increases (left to right). Features driven to zero earliest are weakest linear predictors; those surviving longest are most robust.

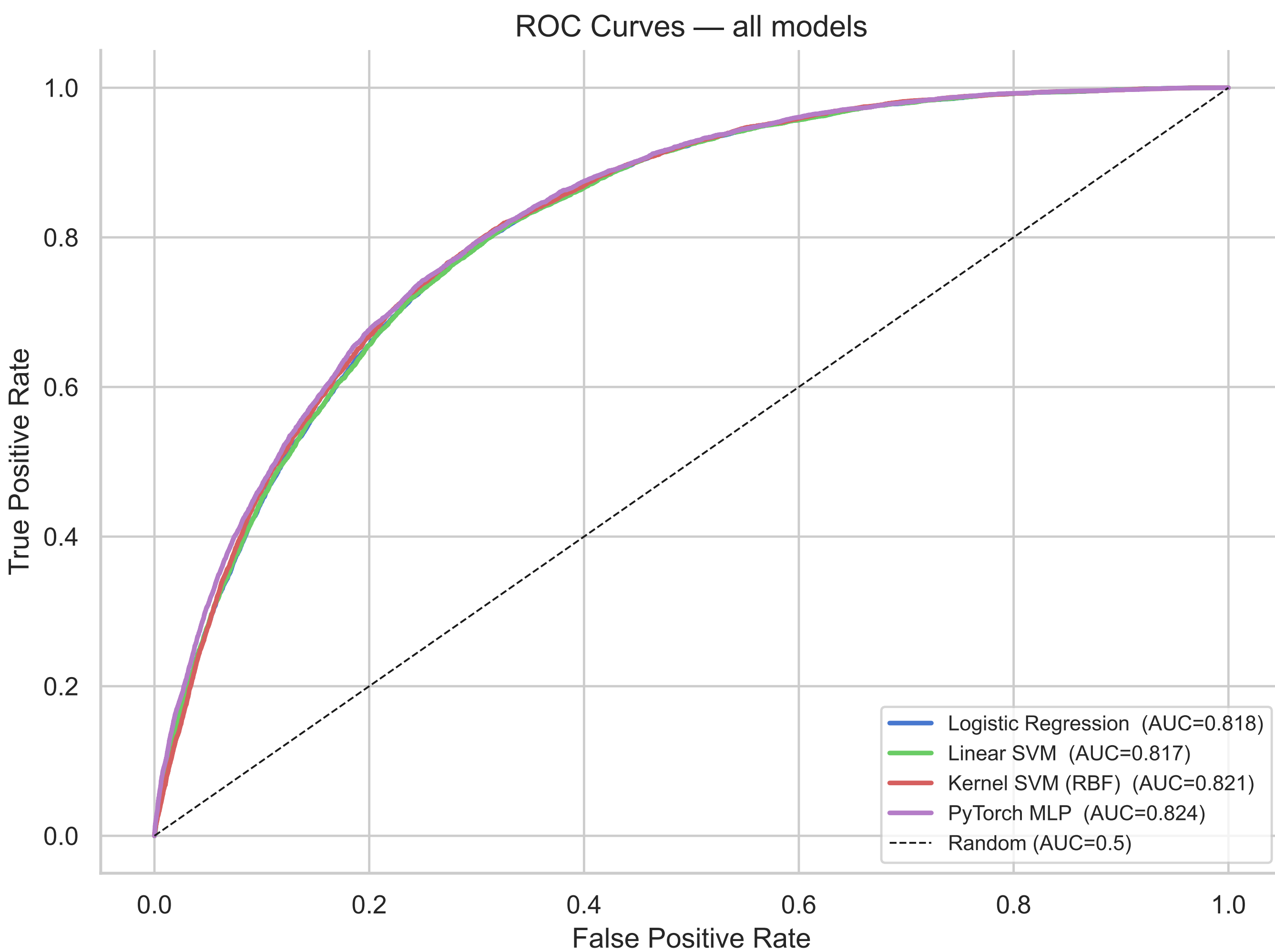

Fig. 18. ROC curves for all four family models; prior to tuning of neural networks.

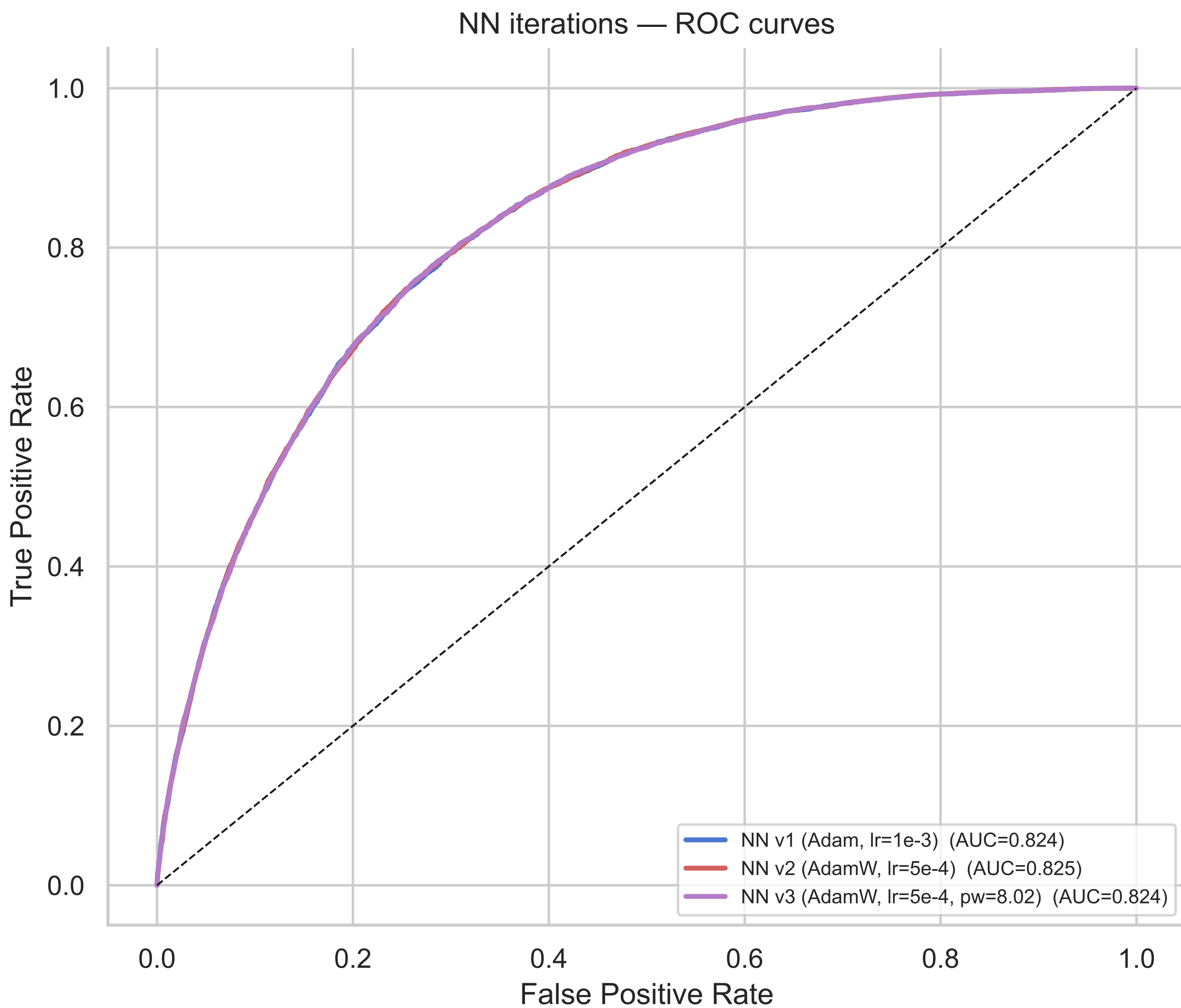

Fig. 19. ROC curves for the three neural networks.